

JACKSONVILLE, IL, USA

WMO: 744666

Lat: 39.780N

Lon: 90.238W

Elev: 624

StdP: 14.37

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 53944

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.3	7.8	-6.4	4.0	4.5	-1.9	5.1	9.2	27.8	23.8	25.0	27.3	7.1	280	0.494

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	92.6	76.2	90.4	75.4	88.2	74.4	79.7	88.6	78.0	87.0	76.5	85.3	8.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.2	144.8	85.4	75.2	135.3	83.7	73.4	127.1	81.7	43.6	88.4	41.9	87.3	40.3	85.8	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.8	20.2	18.1	DB	-5.2	96.4	5.6	3.6	-9.2	99.0	-12.4	101.1	-15.6	103.1	-19.6	105.7	(4)
			WB	-5.9	82.1	5.4	1.5	-9.8	83.1	-13.0	84.0	-16.0	84.8	-20.0	85.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.9	28.2	31.1	43.4	54.4	64.9	73.2	75.8	74.2	67.9	55.4	43.7	33.5
	DBStd	19.24	12.27	11.70	11.50	9.70	8.83	5.98	5.60	5.56	7.66	9.89	10.29	10.13
	HDD50	2370	681	538	272	62	5	0	0	0	1	54	241	516
	HDD65	5226	1142	950	673	337	116	7	1	2	54	329	640	975
	CDD50	3811	5	7	69	194	468	697	800	751	539	222	53	6
	CDD65	1192	0	0	5	19	114	254	336	288	142	33	1	0
	CDH74	11450	0	0	39	228	987	2365	3449	2642	1432	300	8	0
	CDH80	4131	0	0	4	49	283	856	1375	970	514	80	0	0
Wind	WSAvg	8.0	9.7	9.6	10.0	10.1	8.4	7.0	5.4	5.1	5.8	7.5	9.0	8.9
Precipitation	PrecAvg	38.6	1.5	1.5	3.0	4.0	4.8	4.3	4.0	3.5	3.7	2.9	3.1	2.2
	PrecMax	60.0	4.8	4.8	6.5	10.0	10.6	11.7	10.6	7.8	16.0	8.3	10.9	9.6
	PrecMin	25.4	0.0	0.0	0.6	0.6	0.9	0.6	1.0	0.6	0.0	0.3	0.3	0.2
	PrecStd	8.9	1.1	1.0	1.4	2.0	2.5	2.6	2.6	1.9	2.9	1.7	2.2	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.3	67.3	79.1	86.0	90.2	93.1	97.6	96.4	93.5	87.5	74.2	64.4
		MCWB	56.2	55.5	63.6	65.0	71.4	75.6	76.1	77.1	74.2	70.0	59.3	59.6
	2%	DB	55.3	60.2	72.4	80.3	86.4	90.7	92.1	91.4	89.9	81.6	68.7	57.2
		MCWB	50.8	52.7	60.4	61.9	70.0	74.6	78.2	75.9	72.8	66.1	58.2	52.9
	5%	DB	50.1	54.0	66.8	76.0	83.3	88.3	90.1	89.0	86.3	77.0	64.0	52.9
		MCWB	45.2	48.2	57.5	60.9	68.2	73.8	77.4	75.7	71.3	64.3	56.1	48.9
	10%	DB	44.6	47.9	61.5	71.7	79.9	85.5	87.7	85.7	82.1	72.2	59.6	48.3
		MCWB	40.4	42.5	54.6	58.6	66.2	72.3	76.0	74.4	68.8	62.2	52.7	44.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.3	59.2	65.5	67.8	75.1	79.2	82.6	81.5	77.1	73.4	63.8	61.0
		MCDB	62.1	64.4	74.7	79.5	85.8	88.5	89.9	90.0	87.5	84.8	70.3	64.4
	2%	WB	52.2	53.5	61.9	65.4	72.6	77.3	80.3	79.2	75.2	70.7	60.5	54.3
		MCDB	54.6	58.0	70.3	75.4	82.0	86.9	88.6	88.1	85.0	77.9	66.2	56.8
	5%	WB	45.8	48.0	59.2	63.3	70.7	75.7	78.5	77.4	73.6	67.7	57.0	49.3
		MCDB	49.5	54.0	65.5	72.3	79.8	85.5	87.3	86.4	82.4	72.9	62.9	51.8
	10%	WB	40.2	42.7	54.8	60.4	68.6	73.8	77.1	75.5	71.8	64.0	53.3	44.7
		MCDB	44.6	47.7	60.7	69.3	76.9	82.7	85.8	83.0	79.6	70.4	58.7	48.1
Mean Daily Temperature Range	5% DB	MDBR	15.5	16.4	18.6	21.4	19.4	19.1	19.0	19.4	22.9	20.9	17.7	14.1
		MCDBR	22.6	24.5	24.6	27.1	22.1	21.1	20.0	21.8	24.6	25.6	23.1	20.0
		MCWBR	18.0	18.1	14.9	13.9	9.9	9.4	9.2	9.5	10.3	12.5	15.2	16.4
	5% WB	MCDBR	21.4	23.4	22.2	22.9	19.2	18.7	18.2	18.4	19.5	20.0	20.8	19.2
		MCWBR	18.1	18.7	14.4	14.3	9.9	9.4	10.0	9.3	9.0	11.8	16.0	17.1
Clear-Sky Solar Irradiance	taub	0.295	0.307	0.338	0.388	0.425	0.429	0.455	0.445	0.403	0.351	0.303	0.291	
	taud	2.461	2.421	2.415	2.274	2.233	2.264	2.175	2.209	2.294	2.444	2.524	2.488	
	Ebn at Noon	279	290	291	281	272	270	261	261	265	270	274	271	
	Edh at Noon	25	30	34	41	44	43	46	43	37	29	23	23	
All-Sky Solar Radiation	RadAvg	607	872	1177	1537	1750	1994	1988	1749	1500	1034	694	511	
	RadStd	69	82	87	104	136	165	123	107	101	115	63	64	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+189	+135

Nomenclature: See separate page